Finanziato dall'Unione europea NextGenerationEU

Ministero dell'Università e della Ricerca

Italiadomani PIANO NAZIONALE DI RIPRESA E RESILIENZA

Università degli Studi di Ferrara

Does Immigration Affect the Tax Morale of Natives? Evidence from Spain.

Giovanni Cuttica

PhD. student, University of Ferrara

Introduction

Introduction

Immigration is a complex economic and social phenomenon. A lot of research has examined its impact on:

- ➔ the job market (e.g., wages, employment) (Borjas, 2017; Peri, 2014);
- ➔ public finances (more or less expenditure? net fiscal contributors?) (Christl et al., 2022; Fiorio et al., 2023);
- ➔ demand for social services and, in general, the welfare system (Ajzenman et al., 2023; Alesina et al., 2022; Jofre-Monseny et al., 2016; Moriconi et al., 2019)

A less explored aspect is whether immigration affects natives' attitudes toward tax evasion.

Measuring Tax Evasion

Problem: Tax evasion is inherently difficult to observe directly.

Solution: Use *tax morale* as a proxy[1], defined as individuals' intrinsic motivation to comply with tax obligations.

- Lower tax morale $\Rightarrow$ higher likelihood of actual evasion behavior.

Empirical support: tax morale is shaped by both individual socio-demographic factors and broader (local) macroeconomic conditions[2].

social capital, political participation, and positive attitudes toward immigration

vs.

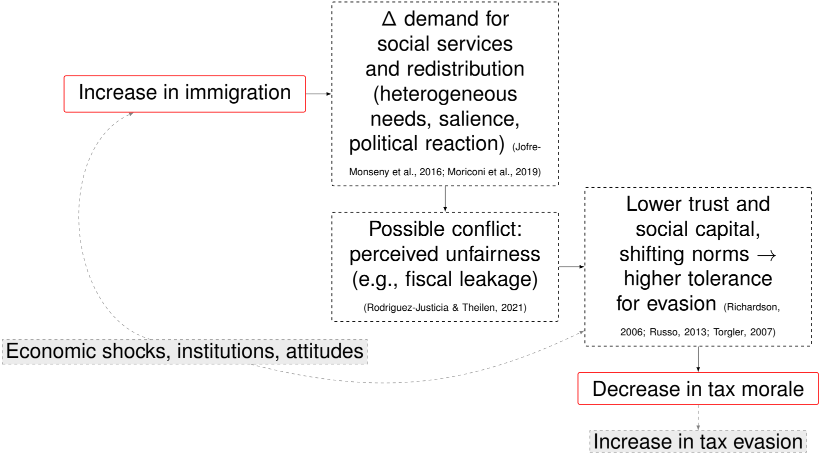
marginalization, dissatisfaction with public services.

Research Question

Does immigration affect tax morale among native citizens?

[1] (Alm, 2018; Torgler, 2007) [2] (María-Dolores et al., 2010; Rodriguez-Justicia & Theilen, 2021; Russo, 2013)

Mechanism


Increase in immigration
Δ demand for social services and redistribution (heterogeneous needs, salience, political reaction) (Jofre-Monseny et al., 2016; Moriconi et al., 2019)
Possible conflict: perceived unfairness (e.g., fiscal leakage) (Rodriguez-Justicia & Theilen, 2021)
Lower trust and social capital, shifting norms → higher tolerance for evasion (Richardson, 2006; Russo, 2013; Torgler, 2007)
Economic shocks, institutions, attitudes
Decrease in tax morale
Increase in tax evasion

Dataset and Aggregation Strategy

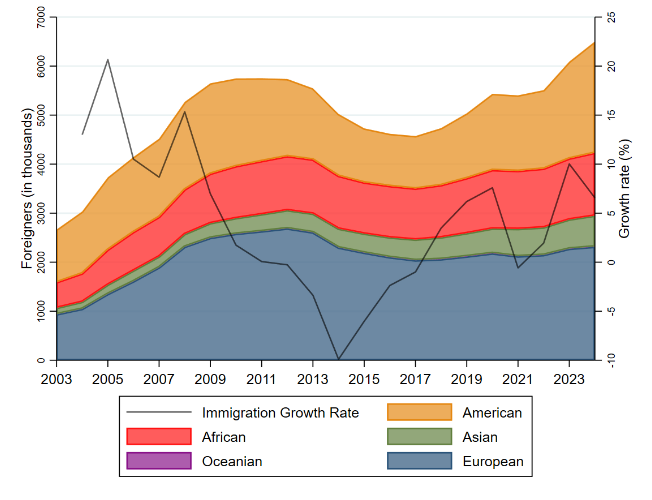
I used data from the Spanish's survey *Opinión Pública y Política Fiscal* (CIS).
2,700 respondents per wave from 2013 to 2024.
The tax morale variable is based on the question:

> *"To what extent do you agree with the following statement:* ***what importance do you give to not evading taxes?****" (0 = "not relevant", 10 = "very important").*

The survey does not report municipalities, although respondents are classified by province and by municipality size (seven categories).

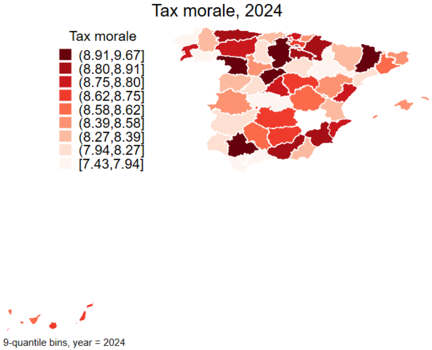
1. Construct territorial units: *municipality size* $\times$ *province* $=$ *locality*.
2. Aggregate immigration data (INE) by *locality*.
3. Merge the two datasets (survey and INE).

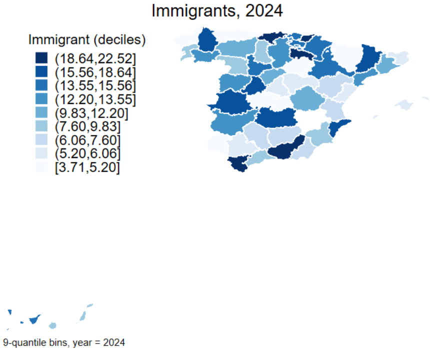
Descriptive Statistics: Immigration Trend



Source: INE and CIS; authors' calculations.

Descriptive Statistics: Tax Morale and Immigration





Empirical Approach

Baseline Model

$$Y_{i,j,t} = \beta \, I_{j,t} + X'_{i,t}\gamma + \lambda_t + \phi_j + \varepsilon_{i,j,t} \tag{1}$$

- $Y_{i,j,t}$: tax morale for individual i in locality j, year t.
- $I_{j,t}$: immigrant share in locality j, year t.
- $X_{i,t}$: individual controls (age, gender, education, type of job).
- λ_t: fixed effects for year (12 years).
- ϕ_j: fixed effects at locality level (220 units).

Interpretation: β is the change in Y for a one-unit increase in the immigrant share. For a 1 p.p. increase, the effect is $0.01 \times \beta$.

Identification concern: time-varying local shocks (labour demand, political climate, institutional quality) may influence both $I_{j,t}$ and $Y_{i,j,t}$, inducing endogeneity.

Shift-Share Instrumental Variable Model

Identification strategy: instrument the immigrant share $I_{j,t}$ using a shift–share design (Borusyak et al., 2025; Card, 2009):

$$Z_{j,t} = \sum_{k \in \mathcal{K}} \underbrace{\left(\frac{\text{Migrant stock}_{j,k,0}}{\text{Population}_{j,0}} \right)}_{s_{j,k}} \cdot \underbrace{\left(\frac{\text{New migrants}_{k,t}}{\text{Migrant stock}_{k,t}} \right)}_{G_{k,t}} = \sum_{k \in \mathcal{K}} s_{j,k}\, G_{k,t}$$

$$\text{for } \mathcal{K} = \{\text{African},\ \text{American},\ \text{Asian},\ \text{European},\ \text{Oceanian}\}. \quad (2)$$

- $s_{j,k}$: historical exposure of locality j to immigrants from origin k (predetermined).
- $G_{k,t}$: national inflow rate for immigrants from origin k at time t (origin–year shock).

$$\textit{First stage:} \quad I_{j,t} = \theta\, Z_{j,t} + \bar{X}'_{j,t}\delta + \lambda_t + \phi_j + \mu_{j,t} \quad (3)$$

$$\textit{Second stage:} \quad Y_{i,j,t} = \beta\, \widehat{I_{j,t}} + X'_{i,t}\gamma + \lambda_t + \phi_j + \varepsilon_{i,j,t} \quad (4)$$

Identification assumptions

$$\text{Exogeneity (shifts): } \mathrm{Cov}\big(Z_{j,t},\ \bar{\varepsilon}_{j,t} \mid \lambda_t, \phi_j, \bar{X}_{j,t}\big) = 0,$$

$$\text{Relevance: } \mathrm{Cov}\big(Z_{j,t},\ I_{j,t} \mid \lambda_t, \phi_j, \bar{X}_{j,t}\big) \neq 0.$$

- $\bar{\varepsilon}_{j,t}$ is the mean residual in cell (j, t).
- Shares baseline $t_0 = 2003$.
- Year and locality fixed effects.
- SEs clustered at locality j level.

Results

Results

Tax Morale	OLS (1)	(2)	IV (3)	(4)
Immigrant share	0.380		1.610**	
	(0.524)		(0.733)	
African share		-5.125***		-5.148***
		(1.090)		(0.996)
American share		5.144**		7.116**
		(2.447)		(3.336)
Asian share		5.436		-0.299
		(3.306)		(4.179)
EU share		-0.055		1.364
		(0.704)		(0.915)
Covariates	✓	✓	✓	✓
Year Fixed Effect	✓	✓	✓	✓
Locality Fixed Effect	✓	✓	✓	✓
Observations	30,312	30,312	30,312	30,312
R-squared	0.035	0.036	0.015	0.017

Clustered standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Source: INE and CIS; authors' calculations.

Discussion of the results

Average effects do not support the hypothesis that immigration reduces natives' tax morale.

However, decomposing immigrant shares by origin reveals heterogeneity.

- ➔ The African share is negatively associated with tax morale in both OLS and IV; the IV estimate is $\hat{\beta} = -5.148$, so a 5 p.p. increase predicts -0.26 points on the 0 to 10 scale.
- ➔ The American share, mostly Latin American immigrants, is positively associated; $\hat{\beta} = 7.116$, so 5 p.p. predicts $+0.36$ points.

The magnitudes are modest, for example $0.26/1.94 \approx 0.13$ sd.

Checking the instrument

Some checks:

- first-stage statistics indicate a strong instrument (Frame 25);
- results are stable to alternative spatial aggregation, whether using province-by-habitat cells or province-level aggregates only (Frame 26);
- dynamic reduced-form estimates show no significant evidence of pre-trends (Frame 27).

Interpretation

The results suggest that composition matters more than volume.

The literature says:

- *cultural proximity* $\uparrow \Rightarrow$ *reciprocity and shared norms* $\uparrow \Rightarrow$ *tax morale* $\uparrow$ (Jeannet, 2020; Torgler, 2007).
- *perceived distance* $\uparrow \Rightarrow$ *fiscal alienation* $\uparrow \Rightarrow$ *tax morale* $\downarrow$ (Richardson, 2006).

Consistent with my estimates, African share $\downarrow$ tax morale, American share $\uparrow$ tax morale, in line with evidence that greater perceived distance lowers support for redistribution and institutional trust (Rodriguez-Justicia & Theilen, 2021; Russo, 2013).

Conclusion

Conclusion

Broader implications

- ➔ Immigrants' positive net fiscal contributions may be partially offset by negative behavioral spillovers among natives .
- ➔ This endogenous channel is often neglected in conventional fiscal impact assessments (Christl et al., 2022; Fiorio et al., 2023).

Caveats: Estimating an IV model with fixed effects and an ordinal outcome is methodologically challenging. I treated tax morale as a continuous variable. However:

- The Linear Probability Model (LPM) may overstate marginal effects near the bounds of the scale.
- Therefore, estimates should be interpreted cautiously: *the direction and significance of effects are more informative than their magnitude*.

Thank you for your attention!

Questions?

Giovanni Cuttica | giovanni.cuttica@unife.it

References I

Ajzenman, N., Dominguez, P., & Undurraga, R. (2023).Immigration, crime, and crime (mis)perceptions. **American Economic Journal: Applied Economics**, **15**(4), 142–176. https://doi.org/10.1257/app.20210156

Alesina, A., Miano, A., & Stantcheva, S. (2022).Immigration and redistribution. **The Review of Economic Studies**, **90**(1), 1–39. https://doi.org/10.1093/restud/rdac011

Alm, J. (2018).What motivates tax compliance? **Journal of Economic Surveys**, **33**(2), 353–388. https://doi.org/10.1111/joes.12272

Borjas, G. J. (2017).The wage impact of the marielitos: A reappraisal. **ILR Review**, **70**(5), 1077–1110. https://doi.org/10.1177/0019793917692945

Borusyak, K., Hull, P., & Jaravel, X. (2025).A practical guide to shift-share instruments. **Journal of Economic Perspectives**, **39**(1), 181–204. https://doi.org/10.1257/jep.20231370

Card, D. (2009).Immigration and inequality. **American Economic Review**, **99**(2), 1–21. https://doi.org/10.1257/aer.99.2.1

Christl, M., Bélanger, A., Conte, A., Mazza, J., & Narazani, E. (2022).Projecting the fiscal impact of immigration in the european union. **Fiscal Studies**, **43**(4), 365–385. https://doi.org/10.1111/1475-5890.12314

Fiorio, C. V., Frattini, T., Riganti, A., & Christl, M. (2023).Migration and public finances in the eu. **International Tax and Public Finance**, **31**(3), 635–684. https://doi.org/10.1007/s10797-023-09787-9

Jeannet, A.-M. (2020).A threat from within? perceptions of immigration in an enlarging european union. **Acta Sociologica**, **63**(4), 343–360. https://doi.org/10.1177/0001699319890652

References II

Jofre-Monseny, J., Sorribas-Navarro, P., & Vázquez-Grenno, J. (2016).Immigration and local spending in social services: Evidence from a massive immigration wave. **International Tax and Public Finance**, **23**(6), 1004–1029. https://doi.org/10.1007/s10797-016-9399-y

María-Dolores, R., Alarcón-García, G., & Garre, M. (2010).Tax morale in spain: A study into some of its principal determinants. **Journal of Economic Issues**, **44**, 855–876. https://doi.org/10.2307/25800724

Moriconi, S., Peri, G., & Turati, R. (2019).Immigration and voting for redistribution: Evidence from european elections. **Labour Economics**, **61**, 101765. https://doi.org/https://doi.org/10.1016/j.labeco.2019.101765

Peri, G. (2014).Do immigrant workers depress the wages of native workers? **IZA World of Labor**, (42). https://doi.org/10.15185/izawol.42

Richardson, G. (2006).Determinants of tax evasion: A cross-country investigation. **Journal of International Accounting, Auditing and Taxation**, **15**(2), 150–169. https://doi.org/10.1016/j.intaccaudtax.2006.08.005

Rodriguez-Justicia, D., & Theilen, B. (2021).Immigration and tax morale: The role of perceptions and prejudices. **Empirical Economics**, **62**(4), 1801–1832. https://doi.org/10.1007/s00181-021-02083-7

Russo, F. F. (2013).Tax morale and tax evasion reports. **Economics Letters**, **121**(1), 110–114. https://doi.org/https://doi.org/10.1016/j.econlet.2013.07.004

Torgler, B. (2007, July). **Tax compliance and tax morale: A theoretical and empirical analysis**. Edward Elgar Publishing. https://doi.org/10.4337/9781847207203

Appendix

Descriptive Stats.

Table: Summary Statistics

Variables	Mean	Std. Dev.	Min	Max
Tax morale score	8.71	1.94	0	10
Age	50.15	17.14	18	99
Male (1 = male)	50.13	50.00	0	1
Higher education (1 = yes)	19.73	39.79	0	1
Low-Skilled Job (1=yes)	12.38	32.93	0	1
Immigrant	10.76	6.13	1.13	39.14
Center-South America	2.77	1.97	0.12	9.84
America (others)	0.31	0.35	0.01	4.40
East Europe	2.22	1.82	0.03	14.19
Europe (others)	2.27	2.89	0.10	23.16
North Africa	1.76	1.76	0.04	20.69
Africa (others)	0.50	0.52	0.01	4.21
Asia	0.92	0.96	0.00	4.76
Oceania	0.01	0.01	0.00	0.28
Total observations (2013-2023)		30,312		

Notes: All immigration variables are expressed in percentage points (%). *Source:* INE and CIS; authors' calculations.

First-stage regressions

Instruments (Z^k)	Total (1)	African (2)	American (3)	Asian (4)	EU (5)
Shift-share Total	40.437*** (2.482)	–	–	–	–
Shift-share African	–	52.298*** (6.384)	1.358 (2.105)	1.135 (0.967)	5.965 (4.702)
Shift-share American	–	-0.281 (1.563)	20.423*** (1.655)	0.835 (0.876)	1.175 (3.989)
Shift-share Asian	–	-13.955 (31.025)	68.683* (36.746)	213.247*** (22.935)	285.811*** (99.233)
Shift-share European	–	-0.929 (0.489)	-0.610 (0.457)	-0.418 (0.316)	17.267*** (1.473)
F-stat excluded instruments	265.38	19.49	72.41	32.50	68.58
SW F-stat (1, clustered)	265.38	74.70	217.69	170.00	126.02
Observations	30,312		30,312		
Fixed Effects	✓		✓		

Notes: Each column reports a separate first-stage regression for an endogenous variable (I^k), instrumented by shift-share variables (Z^k). All regressions include controls: male, age, and education level. Clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.
Source: INE and CIS; authors' calculations.

Go back to frame 16

Different Aggregations

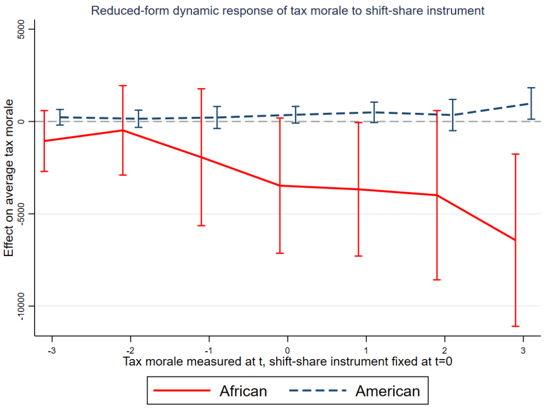
Tax Morale	OLS Localities (3 cat.)		OLS Provinces		IV Localities (3 cat.)		IV Provinces	
Immigrant share	0.352		1.547		1.415**		10.803	
	(0.481)		(3.868)		(0.650)		(10.351)	
African share		-5.052***		-10.808		-5.694***		-46.588*
		(1.401)		(14.976)		(1.221)		(26.456)
American share		3.581		14.348**		1.535		9.328
		(2.818)		(5.722)		(4.455)		(10.493)
European share		1.232		-3.590		1.618		-2.518
		(0.907)		(4.222)		(1.042)		(4.248)
Asian share		-0.932		-21.501		1.214		-37.707
		(3.417)		(28.287)		(4.124)		(45.466)
Covariates	✓	✓	✓	✓	✓	✓	✓	✓
Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
Observations	30,312	30,312	30,312	30,312	30,312	30,312	30,312	30,312
R-squared	0.035	0.035	0.035	0.035	0.016	0.016	0.015	0.015

Clustered standard errors in parentheses *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Note: The 3 categories that I used to aggregate the municipalities based on population at the provincial level are: fewer than 10,000, (2) 10,000–100,000, (3) more than 100,000.
Source: INE and CIS; authors' calculations.

Go back to frame 16

Robustness Check: Time-Structure Diagnostic of the Instrument



Go back to frame 16